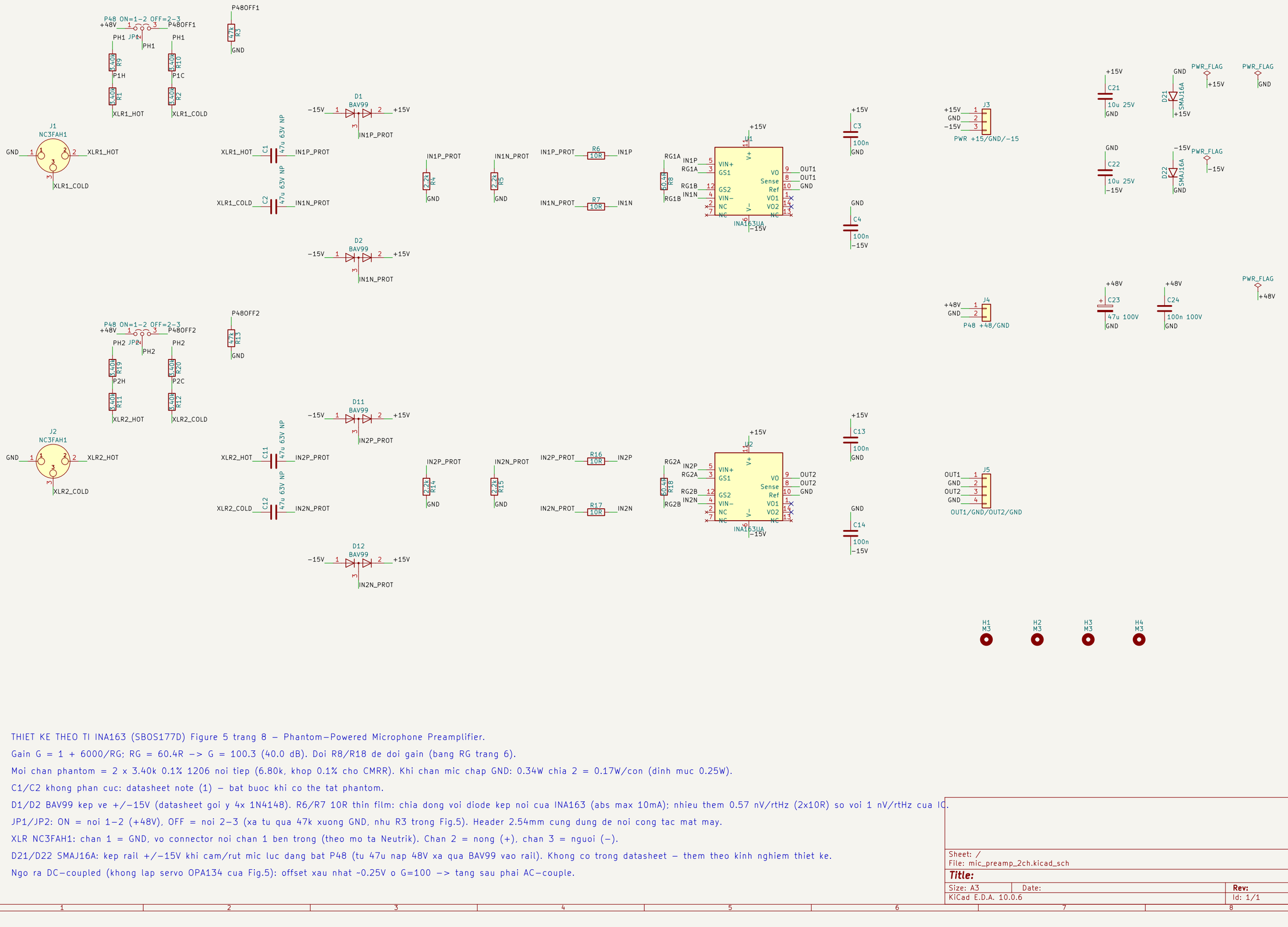


THIET KE THEO TI INA163 (SBOS177D) Figure 5 trang 8 – Phantom-Powered Microphone Preamplifier.

Gain G = 1 + 6000/RG; RG = 60.4R -> G = 100.3 (40.0 dB). Doi R8/R18 de doi gain (bang RG trang 6).

Moi chan phantom = 2 x 3.40k 0.1% 1206 noi tiep (6.80k, khop 0.1% cho CMRR). Khi chan mic chap GND: 0.34W chia 2 = 0.17W/con (dinh muc 0.25W).

C1/C2 khong phan cuc: datasheet note (1) – bat buoc khi co the tat phantom.

D1/D2 BAV99 kep ve +/−15V (datasheet goi y 4x 1N4148). R6/R7 10R thin film: chia dong voi diode kep noi cua INA163 (abs max 10mA); nhieu them 0.57 nV/rtHz (2x10R) so voi 1 nV/rtHz cua IC.

JP1/JP2: ON = noi 1–2 (+48V), OFF = noi 2–3 (xa tu qua 47k xuong GND, nhu R3 trong Fig.5). Header 2.54mm cung dung de noi cong tac mat may.

XLR NC3FAH1: chan 1 = GND, vo connector noi chan 1 ben trong (theo mo ta Neutrik). Chan 2 = nong (+), chan 3 = nguoi (−).

D21/D22 SMAJ16A: kep rail +/−15V khi cam/rut mic luc dang bat P48 (tu 47u nap 48V xa qua BAV99 vao rail). Khong co trong datasheet – them theo kinh nghiem thiet ke.

Ngo ra DC-coupled (khong lap servo OPA134 cua Fig.5): offset xau nhat ~0.25V o G=100 -> tang sau phai AC-couple.

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